



KBCC

Boron Carbide Composite

KEY PROPERTIES

- Extremely high hardness
- Low density
- Suitable for EDM machining
- High wear resistance

APPLICATIONS

- Cutting tools
- Watch-making industry
- Wear resistant parts

MAIN PROPERTIES

Property	Value	Unit
Density	3.0	g/cm ³
Flexural strength	635	MPa
Hardness	32	GPa
Fracture Toughness	3.5	MPa·√m

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		800	°C
Recomended temp		<750	°C
Continuous temp		700	°C

ELECTRICAL BEHAVIOR

Semi-conductive

MATERIAL INFORMATION

Material type		Boron Carbide composite
Structure		Dense ceramic
Porosity level		EDM machinable
Composition		Ultra-high hardness
Color		Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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